```
#include<iostream>
using namespace std;
const int MAX=5;

class queue
{
   private:
   int a[MAX];
   int front;
   int rear;

   public:
   queue()
   {
      front = -1;
      rear = -1;
   }

   int empty(void)
   {
      if (front == -1 && rear == -1)
      {
         return 1;
      }
      else
      {
         return 0;
      }
   }

   int full(void)
   {
      if (rear == MAX - 1)
      {
         return 1;
      }
      else
      {
         return 0;
      }
   }

   void enqueue(int x)
   {
      if (full())
      {
         cout << "\nQueue is full";
         return;
```

```
    }

    if (front == -1 && rear == -1)
    {
        front++;
        rear++;
    }
    else
    {
        rear++;
    }

    a[rear] = x;
}

int dequeue(void)
{
    if (empty())
    {
        cout << "\nQueue is empty";
        return -1;
    }

    int x = a[front];

    if (front == rear)
    {
        front = -1;
        rear = -1;
    }
    else
    {
        front++;
    }

    return x;
}

int peek(void)
{
    if (empty())
    {
        cout << "\nQueue is empty";
        return -1;
    }

    return a[front];
}
```

```
    void display(void)
    {
        if (empty())
        {
            cout << "\nQueue is empty";
            return;
        }

        cout << "\nQueue elements:";
        for (int i = front; i <= rear; i++)
        {
            cout << "\t" << a[i];
        }
    }
};

int main()
{
    int x;
    queue q;

    q.enqueue(50);
    q.enqueue(18);
    q.enqueue(27);
    q.enqueue(35);
    q.enqueue(89);

    q.display();

    if (!q.empty())
    {
        x = q.dequeue();
        cout << "\nRemoved: " << x;
    }

    q.display();

    x = q.peek();
    cout << "\nFront element is: " << x;

    return 0;

}
//[Running] cd "c:\Users\STUDENT\Desktop\manasi kulkarni 209\" && g++
queueusingarray.cpp -o queueusingarray && "c:\Users\STUDENT\Desktop\manasi kulkarni
209\"queueusingarray
```

```
//Queue elements:     50      18      27      35      89
//Removed: 50
//Queue elements:     18      27      35      89
//Front element is: 18
//[Done] exited with code=0 in 0.591 seconds
```